

Class 13: Partitioned Matrices

In many applications the systems to be solved may involve millions of variables and millions of equations. The matrices are huge! We need to be clever and creative to find shortcuts in doing the computations with these matrices. Sometimes the variables and equations can be grouped together so the matrix is broken up into smaller blocks. If some of these blocks are all zeros or maybe an identity matrix block we can do the large computations more efficiently.

Example: $A = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

Find A^{-1} .

How to multiply block matrices A and B

Key idea: The way you partition the columns of A and the rows of B must be the same!

$$\text{Suppose } A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 4 & 5 & 6 & 7 \\ 3 & 4 & 5 & 6 & 7 & 8 \end{bmatrix}$$

$$\text{and } B = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \\ 5 & 6 & 7 & 8 \\ 6 & 7 & 8 & 9 \end{bmatrix}$$

$$AB = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix}$$

Practice: Suppose A has the form $\begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}$

where A_{11} is square $p \times p$ and A_{22} is square $q \times q$ and suppose A is invertible. Show that both A_{11} and A_{22} must be invertible.

Practice: Suppose A has the form $\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$

where A_{11} is invertible. Find X , Y , and S so that

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} I & 0 \\ X & I \end{bmatrix} \begin{bmatrix} A_{11} & 0 \\ 0 & S \end{bmatrix} \begin{bmatrix} I & Y \\ 0 & I \end{bmatrix}$$